

Edward García-Hernández' Résumé

Personal Information

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Citizenship Colombian

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Education

Universidad del Quindío BSc Physics (2019 - 2024)

- Supervisor: Dr. Édgar Arturo Gómez González
- Co-supervisor: Dr. Stefanie Czischek (University of Ottawa)
- Thesis: "Reconstruction of many-body quantum dynamics from neural networks trained on classical shadows"

University of Ottawa MSc Physics (January 2025 - September 2025 [Fast-Tracked])

- Supervisor: Dr. Stefanie Czischek
- Thesis: Solving the k -charge removal problem using Rydberg atom arrays

University of Ottawa PhD Physics (September 2025 - Present)

- Supervisor: Dr. Stefanie Czischek
- Thesis: Analysis of optimization landscapes in physics problems

Research

Universidad del Quindío (4 months, Part-time) (2021)

Research assistant, Instituto Interdisciplinario de las Ciencias (ICC)

- Established optimal geometrical arrangement for an RGB digital holography setup.

Universidad del Quindío (4 months, Part-time) (2022)

Research assistant, Theoretical and Computational Physics Group (GFTC)

- Calculated properties of a simple Dicke-model Hamiltonian under parity deformation of the defining relation of the bosonic mode creation and annihilation operators.

University of Ottawa (3 months, Full-time) (2023)
Mitacs Globalink Research Intern, APRIQuOt Group

- Coded recurrent neural networks trained on “snapshots” of the density matrix and analyzed their accuracy to predict the dynamics of simple many-body quantum systems.

University of Ottawa/Pasqal (4 months, Full-time) (Winter 2025)
Mitacs Accelerate Research Intern, APRIQuOt Group

- Derived QUBO formulation of the k -charge removal problem.

University of Ottawa/Pasqal (4 months, Full-time) (Summer 2025)
Mitacs Accelerate Research Intern, APRIQuOt Group

- Mapping of the QUBO formulation of the k -charge removal problem onto the neutral atom platform, and finding its solution using the adiabatic approximation.

University of Ottawa/Pasqal (4 months, Full-time) (Fall 2025)
Mitacs Accelerate Research Intern, APRIQuOt Group

- Work on embedding strategy for signed QUBOs using spectral embeddings, SVD, and Schrieffer-Wolf perturbation theory.

University of Ottawa/Pasqal (4 months, Full-time) (Winter 2026)
Mitacs Accelerate Research Intern, APRIQuOt Group

- Spectral characterization of the Crystal Structure Prediction QUBO.

Research interests

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|----------------------------------|---|
| • Cavity Quantum Electrodynamics | • Interpretability of Machine Learning Techniques for Physics |
| • Condensed Matter Physics | • Neutral atom quantum computing |
| • AI for Quantum | • Crystal structure prediction |

Work experience

Universidad del Quindío - Program of Physics (2019 - 2023)
Teaching Assistant

- Mecánica: 1st year elementary newtonian mechanics (1 term)
- Mecánica Cuántica I: 3rd year quantum mechanics (3 terms)

Colegio Gimnasio Contemporáneo (2024)
Teacher

- Substitute teacher (January)
- Main teacher: Geometry 6th grade, Statistics 6th grade, Geometry 7th grade, Statistics 10th grade (February - June)

Outlier.ai (2024, freelance)
AI Tutor

- Testing LLM's responses to undergraduate- and graduate-level math prompts.

University of Ottawa - Department of Physics (2025)
Teaching Assistant

- PHY2323: Electricity and Magnetism (Summer 2025)

University of Ottawa - Department of Physics (2025)
Teaching Assistant

- PHY1321: Principles of Physics I (Fall 2025)

University of Ottawa - Department of Physics (2026)
Teaching Assistant

- PHY1122: Fundamentals of Physics II (Winter 2026)

Coding

Python (Intermediate)
Used for scientific computing

- Scipy, Numpy, PyTorch, Pandas, Matplotlib, Seaborn, JAX

Languages

Spanish (Native Speaker)

English (Full professional proficiency)

- EF-SET $\rightarrow$ 84/100 C2 Proficient. (2022)
- TOEFL-iBT $\rightarrow$ 110/120 Advanced (2024)

French (Elementary proficiency)

Events & Awards

XXVIII Congreso Nacional de Física (National Physics Conference) (2019)

- *Paid assistant fees as a 1st year student due to outstanding academic performance*

Qiskit Global Summer School: Quantum Simulation (2022)

- *Quantum Excellence Badge*

XXIX Congreso Nacional de Física (National Physics Conference) (2022)

- *Poster presentation. Title: “Revisión del proceso de simulación de modelos de espín en computadores cuánticos ruidosos usando Qiskit”*

ICTP’s 10th International Symposium of Optics and its Applications (2022)

- *Representative of GFTC*

Challenges in Simulating Quantum Matter (2025)

- *Poster presentation. Title: “Leveraging Neutral Atom Quantum Computing for Crystal Structure Prediction”*

Canadian Association of Physicists Congress (2026)

- *Poster presentation. Title: “Singular Value Gadgets for Embedding QUBOs on Rydberg Atom Hardware”*